Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

1 Tomato

1.1 Seed Selection and Nursery

- **Seed selection:** For 1 ha of cultivation, 100 g of hybrid tomato (*Solanum lycopersicum*) seeds is required. Recommended varieties or hybrids, seed sources, certification, seed quality standards, germination percentage, viability, vigour, purity, and seed health parameters are Not specified in the provided sources.
- **Seed treatment:** Tomato seeds are treated by mixing with bio-fertilizer *Azospirillum* at 200 g/kg of seed and shade dried for 30 minutes. Prior to transplanting, seedlings are dipped in a solution of 15 ml of Nuvacron and 25 g of Dithane M-45 in 10 L of water for 5–6 minutes. Other seed treatment methods, active ingredient concentrations, durations, and precautions are Not specified in the provided sources.
- **Nursery site and type:** Seedlings are produced in a protected nursery covered with 50 % shade net, using protrays placed on raised beds. Site selection criteria, sunlight exposure, ventilation requirements, drainage systems, and water availability standards are Not specified in the provided sources.
- **Nursery preparation:** Raised beds are formed with a width of 1 m and a convenient length. The growing media consists of sterilized cocopeat at 300 kg/ha mixed with 5 kg of neem cake, along with *Azospirillum* at 1 kg and *Phosphobacteria* at 1 kg. Additional soil preparation steps, media sterilization protocols, sanitation measures, and nursery hygiene details are Not specified in the provided sources.
- **Sowing procedures:** Seeds are sown in protrays using the prepared media at a seed rate of 100 g of hybrid seeds per hectare. After sowing, watering is done using a rose can twice a day up to seed germination. Sowing depth, cell spacing, covering materials, labeling procedures, and exact sowing dates are Not specified in the provided sources.
- **Environmental requirements:** Environmental protection is maintained using a 50 % shade net structure over the nursery area and regular watering. Numerical thresholds for temperature, relative humidity, sunlight intensity, and other environmental conditions for germination and seedling growth are Not specified in the provided sources.
- **Nursery management:** Irrigation is provided twice daily with a rose can up to seed germination. Nutrient management includes spraying Panchakavya at 3 % (30 ml/L) on the 15th day after sowing and drenching with a 19:19:19 + micronutrient solution at 0.5 % (5 g/L) on the 18th day after sowing. Weeding, thinning, shading

adjustments, ventilation controls, and specific pest and disease control measures in the nursery are Not specified in the provided sources.

- **Seedling quality:** Tomato seedlings reach transplanting readiness at 25–30 days old. Germination assessment criteria, seedling height, leaf development stages, stem strength, root growth characteristics, and plant uniformity measurements are Not specified in the provided sources.
- **Hardening and transportation:** Seedlings are transplanted at 25–30 days old. Hardening procedures, timing, seedling removal techniques, root protection methods, handling guidelines, and transportation conditions before transplanting are Not specified in the provided sources.
- **Scheduling and monitoring:** The documented nursery schedule requires placing germinated protrays on raised beds inside the shade net after 6 days, applying Panchakavya spray at 15 days after sowing, drenching 19:19:19 + micronutrient solution at 18 days after sowing, and transplanting 25–30 days old seedlings, preferably in the evening following a 5–6 minutes protective dip. Germination monitoring protocols, recordkeeping systems, and formal follow-up procedures are Not specified in the provided sources.
- **Effects on crop performance:** Dipping seedlings in Nuvacron (15 ml) and Dithane M-45 (25 g) in 10 L of water for 5–6 minutes before planting protects young plants during initial field establishment. Other documented relationships between seed quality, nursery management practices, seedling health, crop establishment, and yield are Not specified in the provided sources.
- **Integrated workflow:** The integrated sequence begins with seed selection at 100 g/ha of hybrid seed, followed by seed treatment with *Azospirillum* at 200 g/kg of seed and shade drying for 30 minutes. Media preparation involves mixing 300 kg/ha of sterilized cocopeat with 5 kg of neem cake, 1 kg of *Azospirillum*, and 1 kg of *Phosphobacteria*, which is filled into protrays on 1 m wide raised beds inside a 50 % shade net structure. Protrays are watered twice daily with a rose can until germination, then placed individually on raised beds at day 6. Nursery care includes Panchakavya spray at 3 % on day 15 and 19:19:19 + micronutrient drenching at 0.5 % on day 18. Seedlings at 25–30 days old are dipped in a solution of Nuvacron (15 ml) and Dithane M-45 (25 g) in 10 L of water for 5–6 minutes and transplanted into the main field, preferably in the evening.